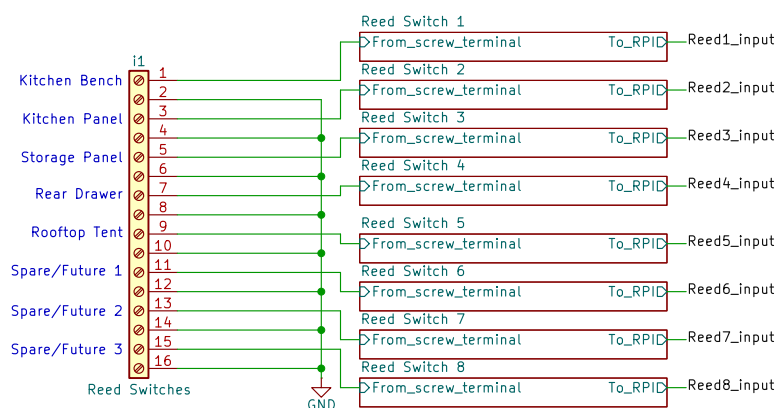
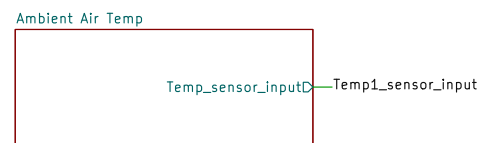


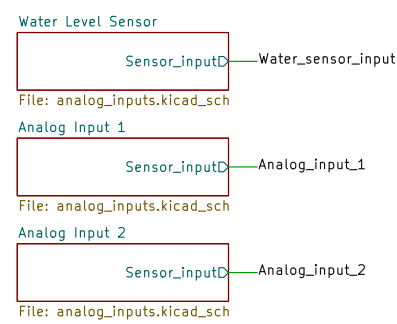
Reed Switches



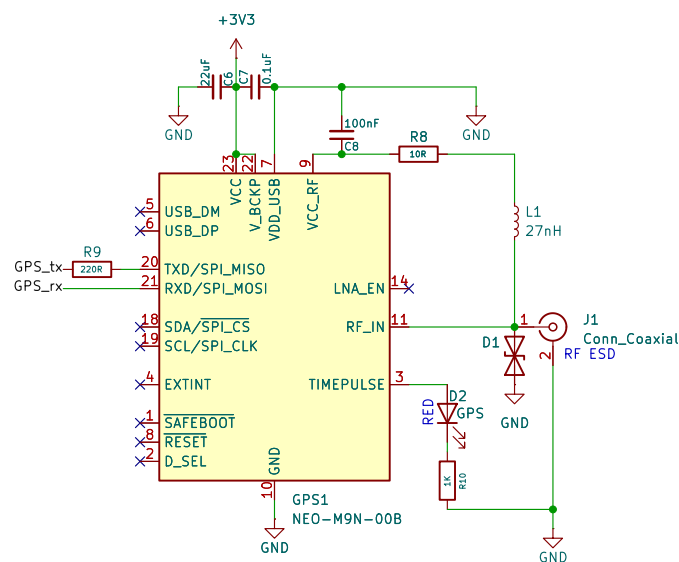
Temp Sensor 1-Wire Buses



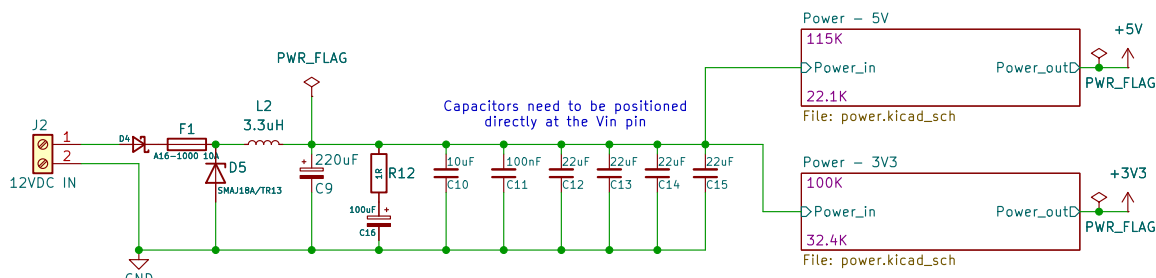
Arduino Analog Inputs



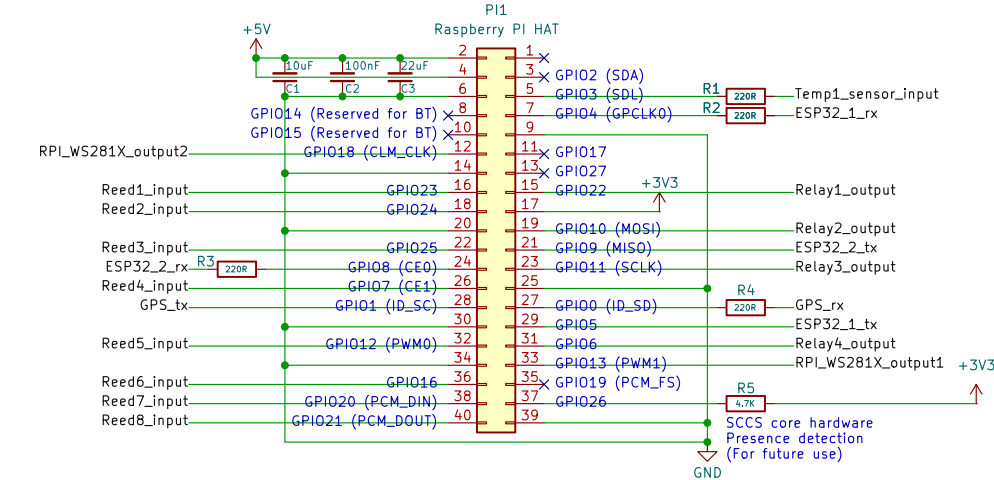
GPS



Power



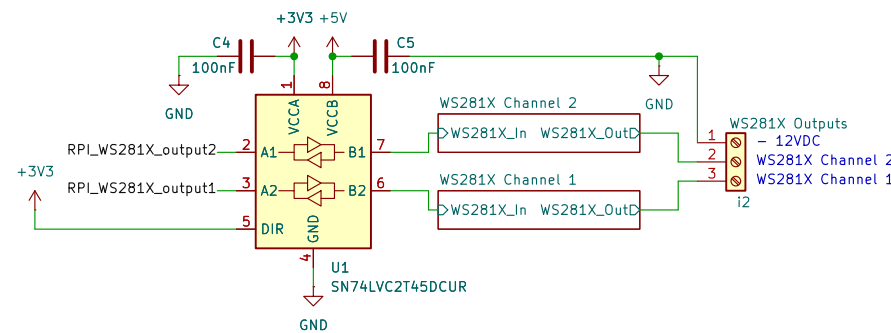
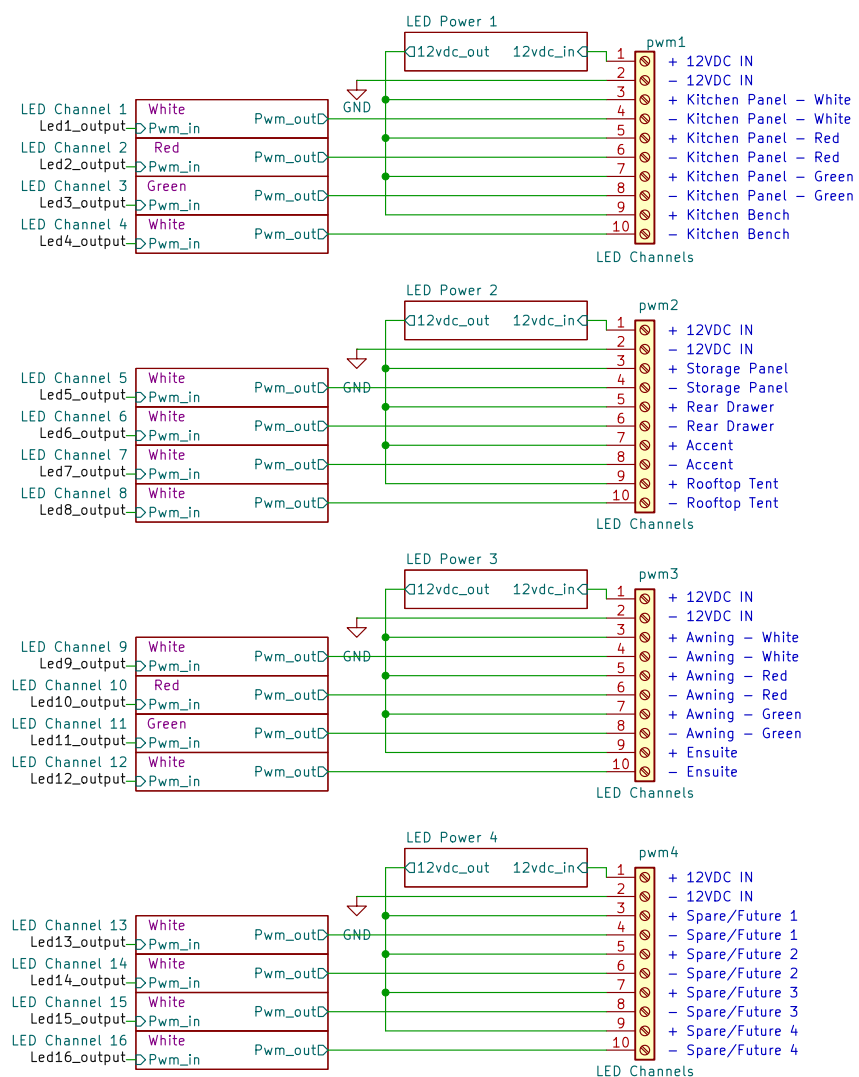
Raspberry Pi



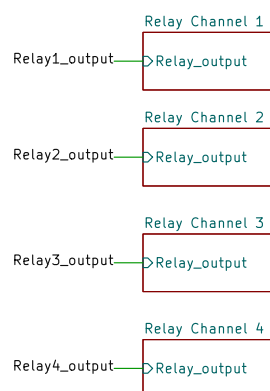
ESP32's



Dimmable Lighting

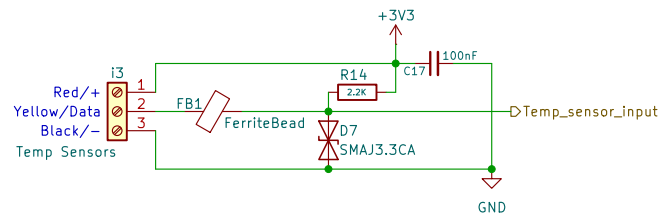


Relay Channels



Notes

- Power circuit provides 5V
- 12V buck (U7 AP64501) provides system 3.3V; PI 3V3 pins isolated
- 1-Wire Bus Notes:
Board has a 2.2K pull-up on the data line.
Put a 220R resistor on the data pin of each temp sensor but ensure it's not in series with the bus.
Put a 100nF ceramic cap across VCC and GND at each temp sensor.
- WS281X RPI Notes:
Put a 220uF electrolytic cap between 12V and GND at the start of each LED strip
- UARTS:
UART1 / GPIO14/15 / Reserved for Bluetooth
UART2 / GPIO00/01 / GPS
UART3 / GPIO04/05 / ESP32 No. 1
UART4 / GPIO08/09 / ESP32 No. 2
- 12V LED Strip Notes (not required but good practice):
Put a 220uF 25V electrolytic cap across 12V and GND at each LED strip.
Put a 100nF 25V ceramic cap across 12V and GND at each LED strip.



Sheet: /Ambient Air Temp/
File: 1_wire_bus.kicad_sch

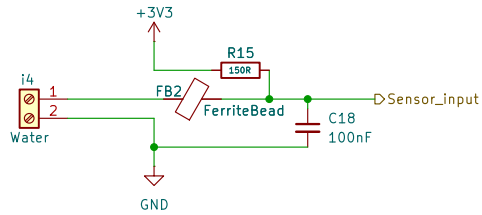
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KiCad E.D.A. 10.0.4

Date:

Rev:

Id: 2/43



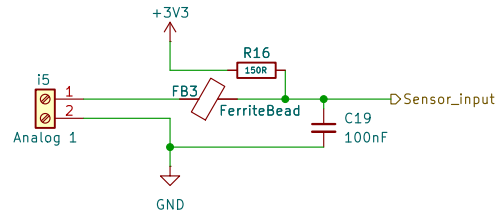
Sheet: /Water Level Sensor/
File: analog_inputs.kicad_sch

Title:

Size: A5
KiCad E.D.A. 10.0.4

Date:

Rev:
Id: 3/43



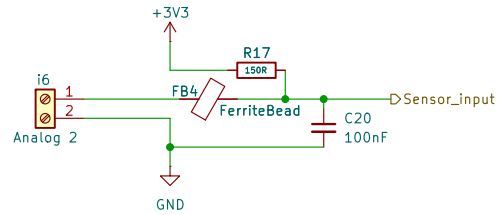
Sheet: /Analog Input 1/
File: analog_inputs.kicad_sch

Title:

Size: A5
KiCad E.D.A. 10.0.4

Date:

Rev:
Id: 4/43



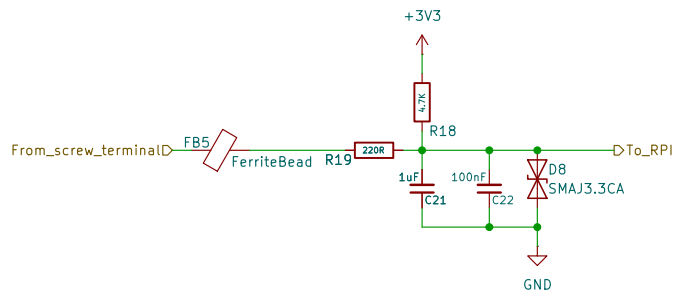
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Date:

Rev:
Id: 5/43



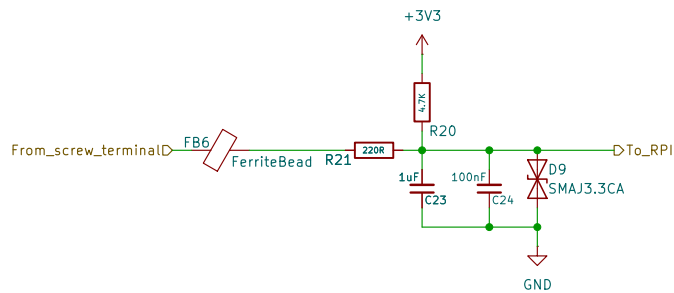
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File: reed_input.kicad_sch

Title:

Size: A5
KiCad E.D.A. 10.0.4

Date:

Rev:
Id: 6/43



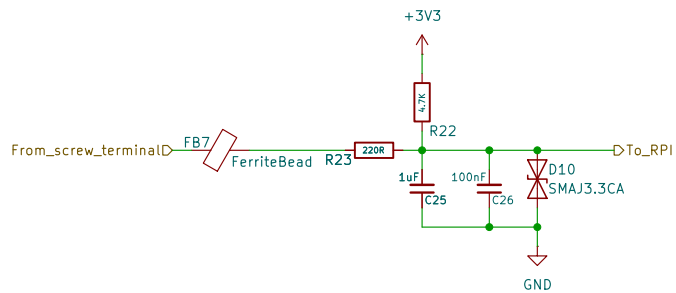
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Date:

Rev:
Id: 7/43



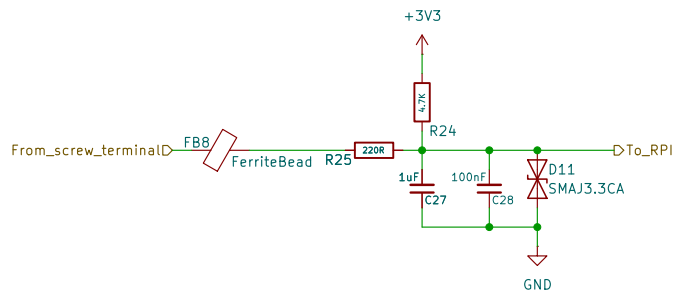
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File: reed_input.kicad_sch

Title:

Size: A5
KiCad E.D.A. 10.0.4

Date:

Rev:
Id: 8/43



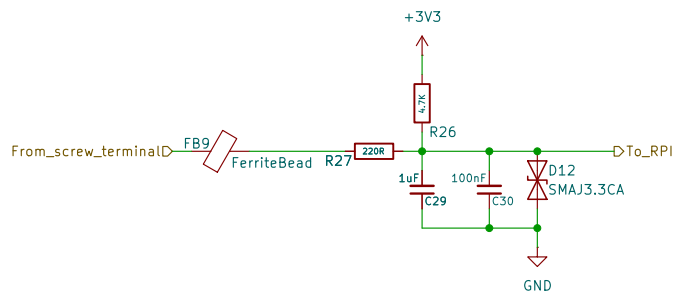
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KiCad E.D.A. 10.0.4

Date:

Rev:
Id: 9/43



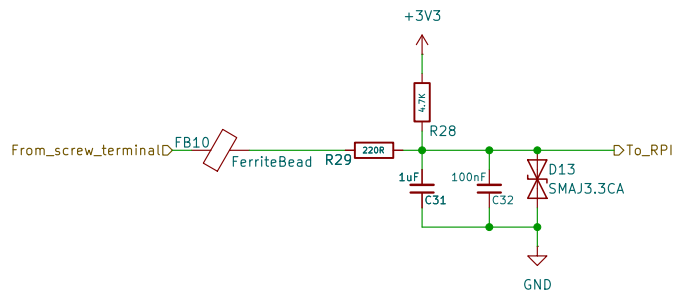
Sheet: /Reed Switch 5/
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Size: A5
KiCad E.D.A. 10.0.4

Date:

Rev:
Id: 10/43



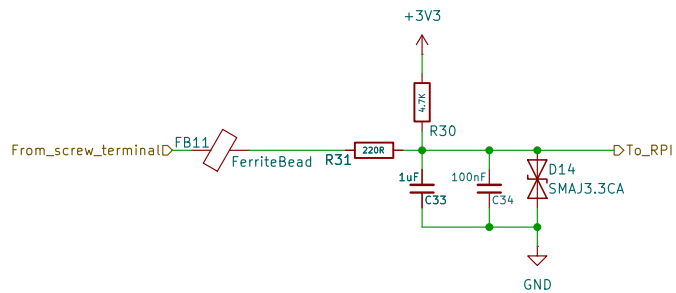
Sheet: /Reed Switch 6/
File: reed_input.kicad_sch

Title:

Size: A5
KiCad E.D.A. 10.0.4

Date:

Rev:
Id: 11/43



Sheet: /Reed Switch 7/
File: reed_input.kicad_sch

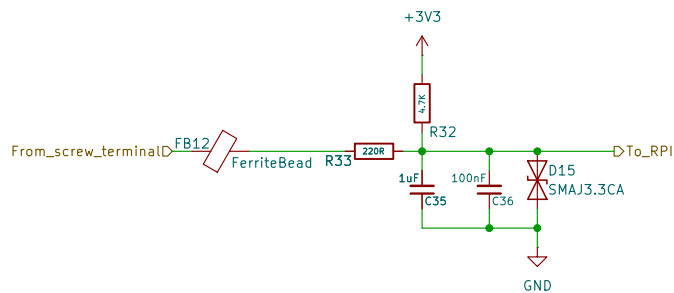
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Rev:

Id: 12/43



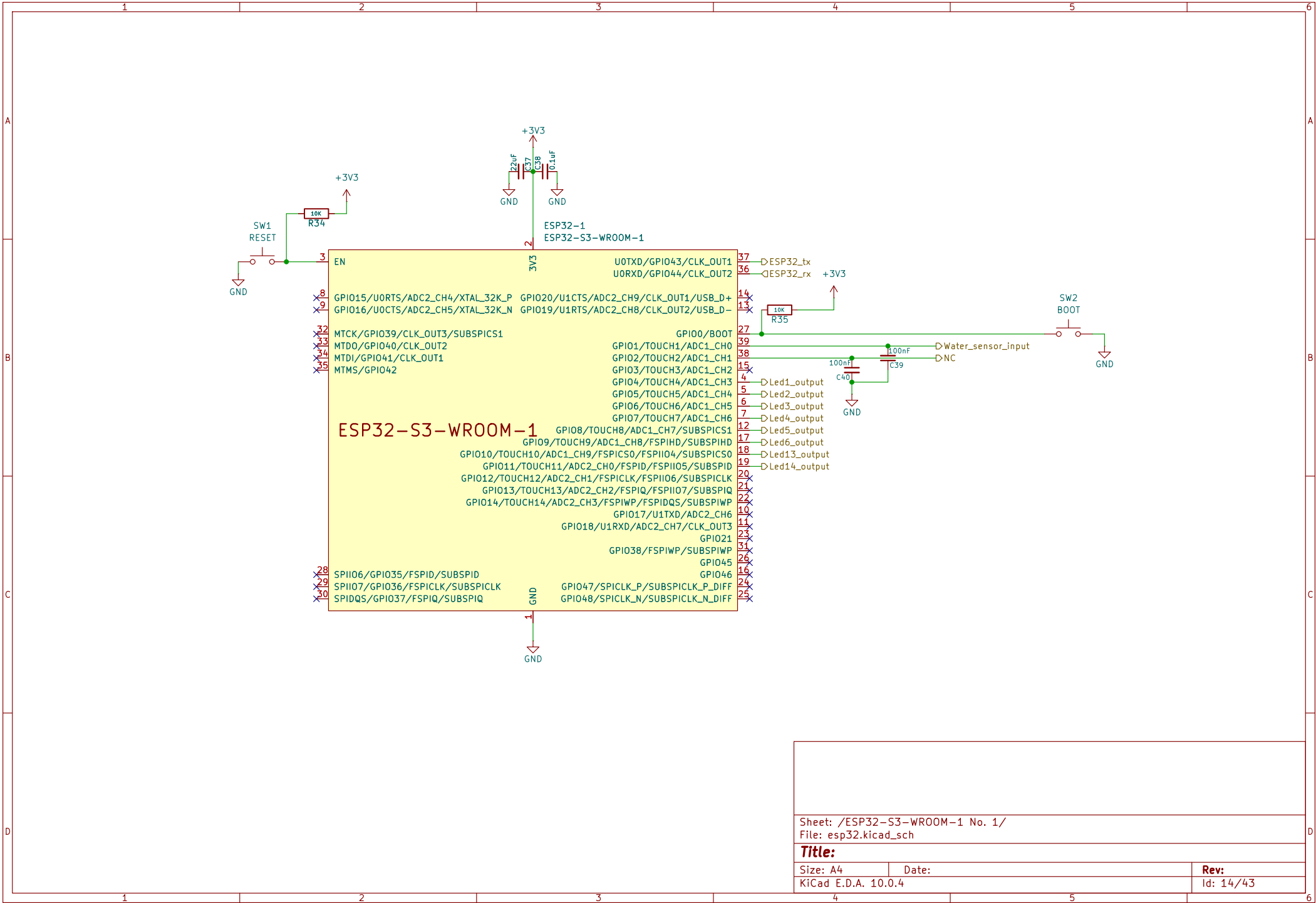
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File: reed_input.kicad_sch

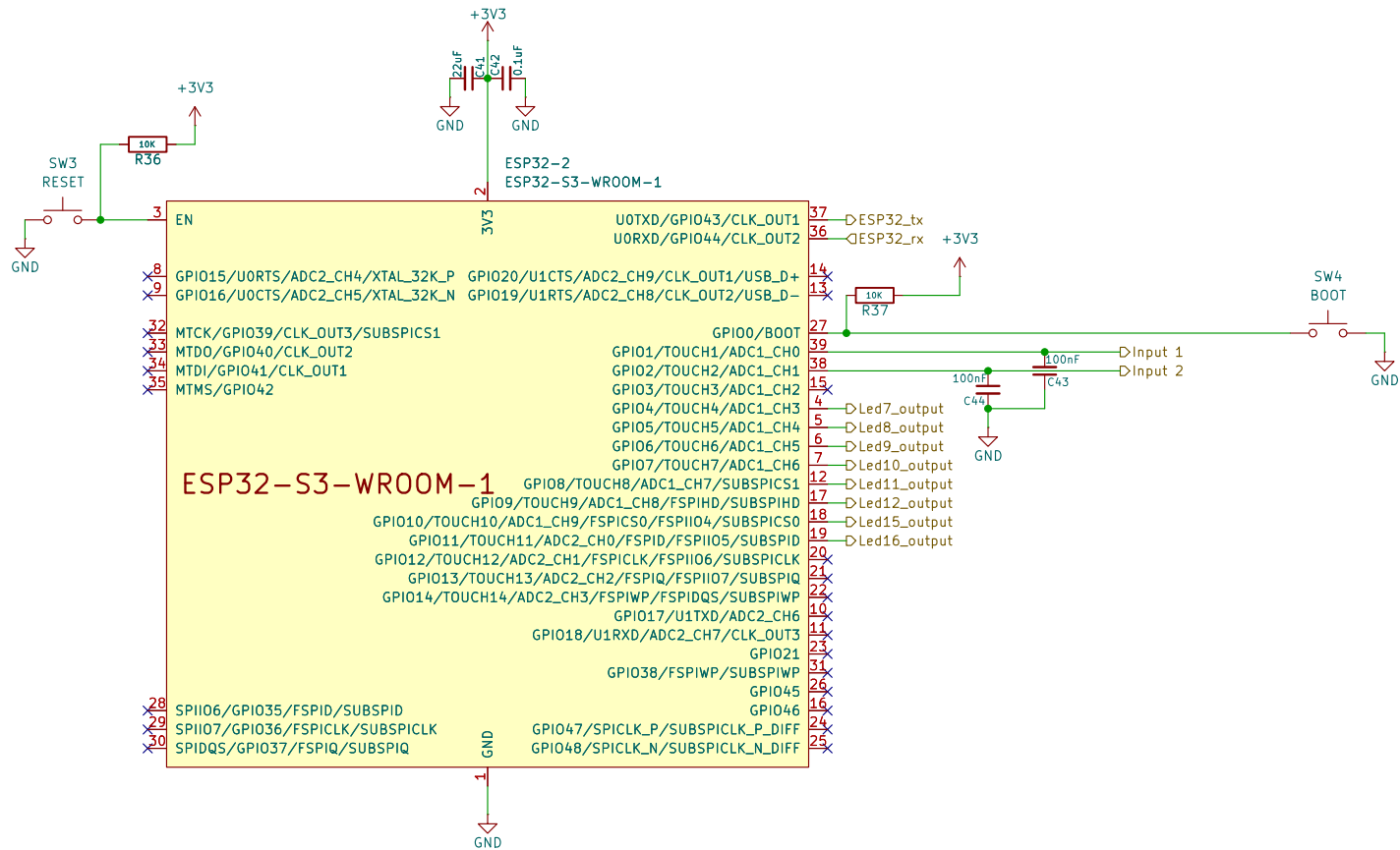
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Date:

Rev:
Id: 13/43





Sheet: /ESP32-S3-WROOM-1 No. 2/
File: esp32.kicad_sch

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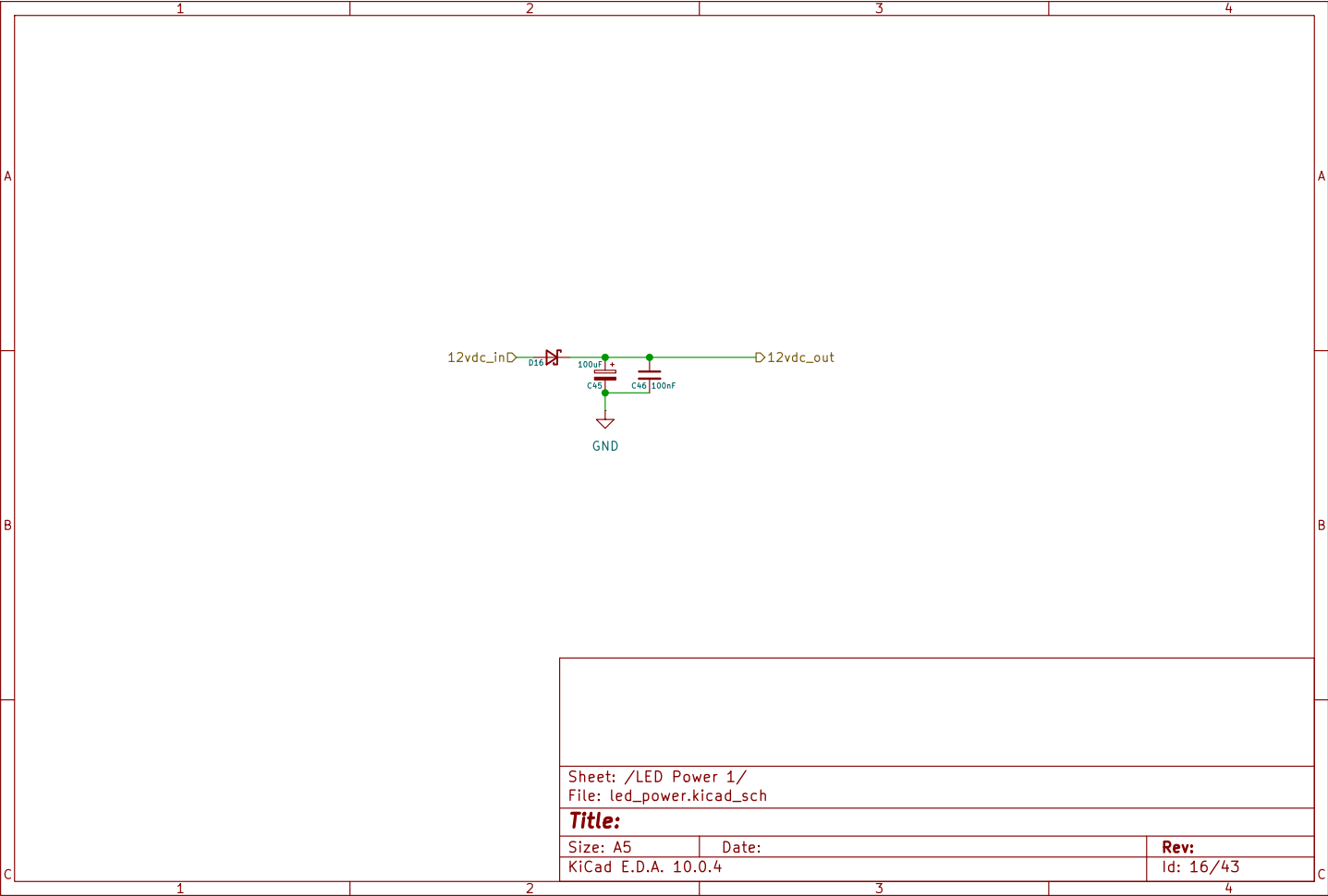
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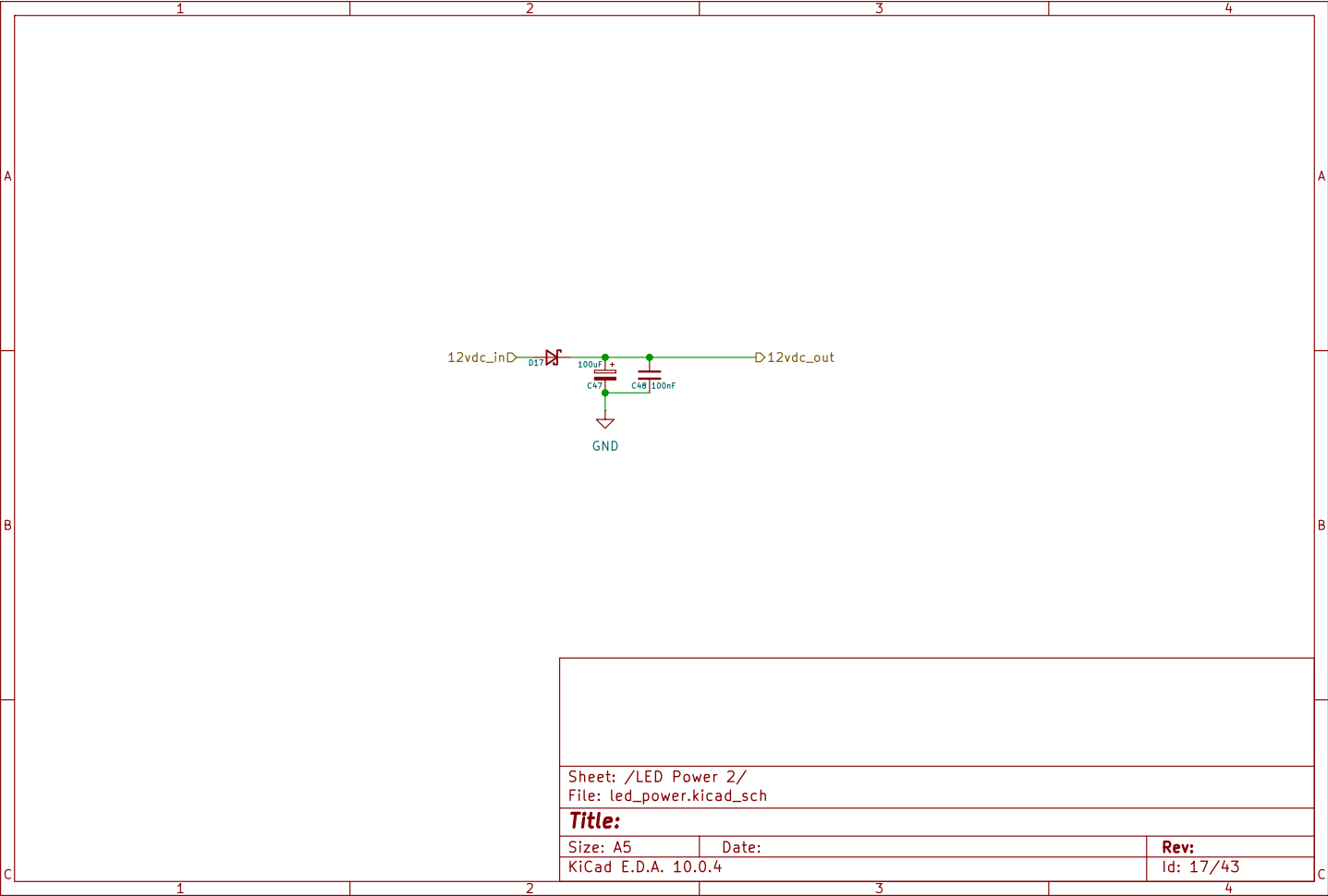
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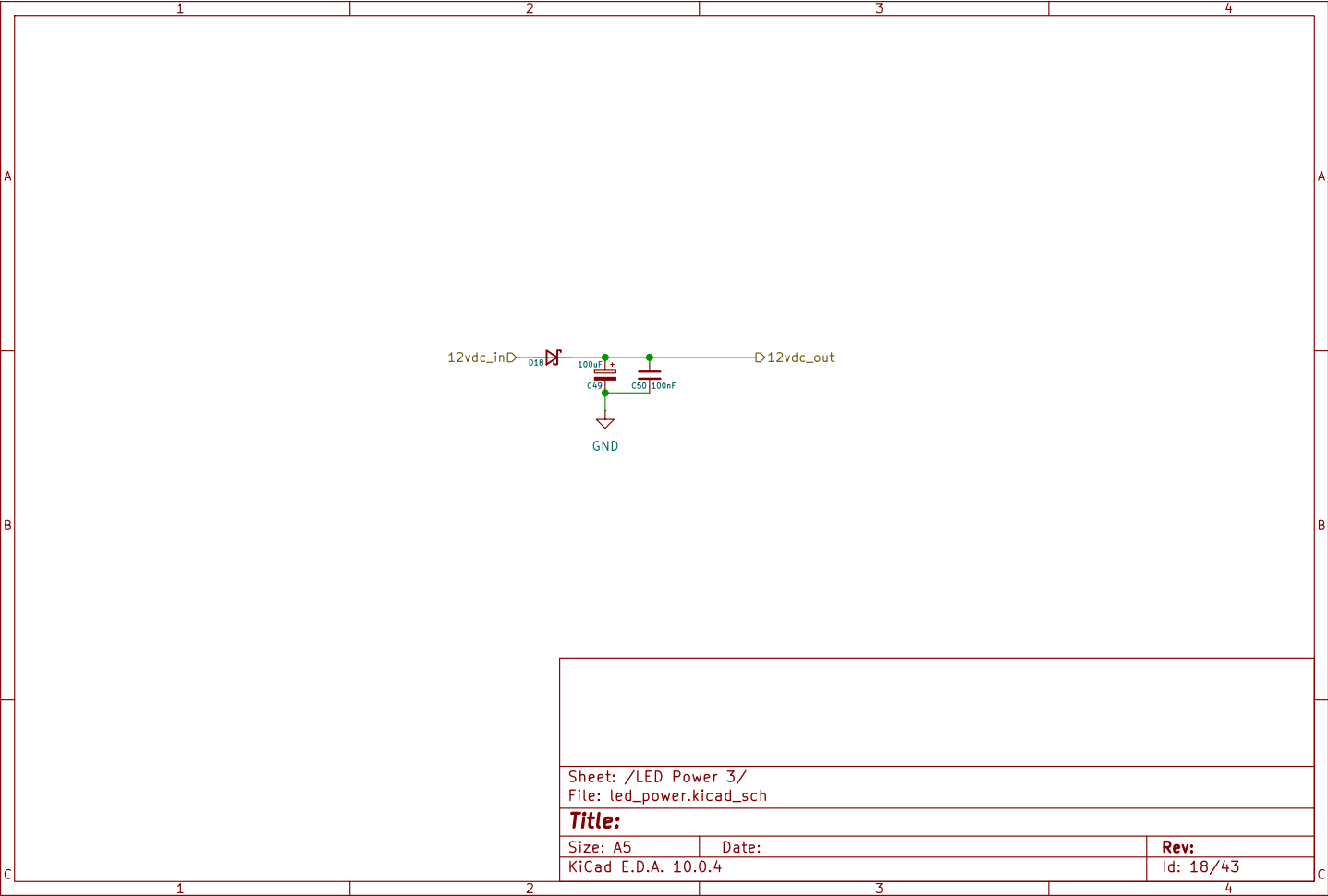
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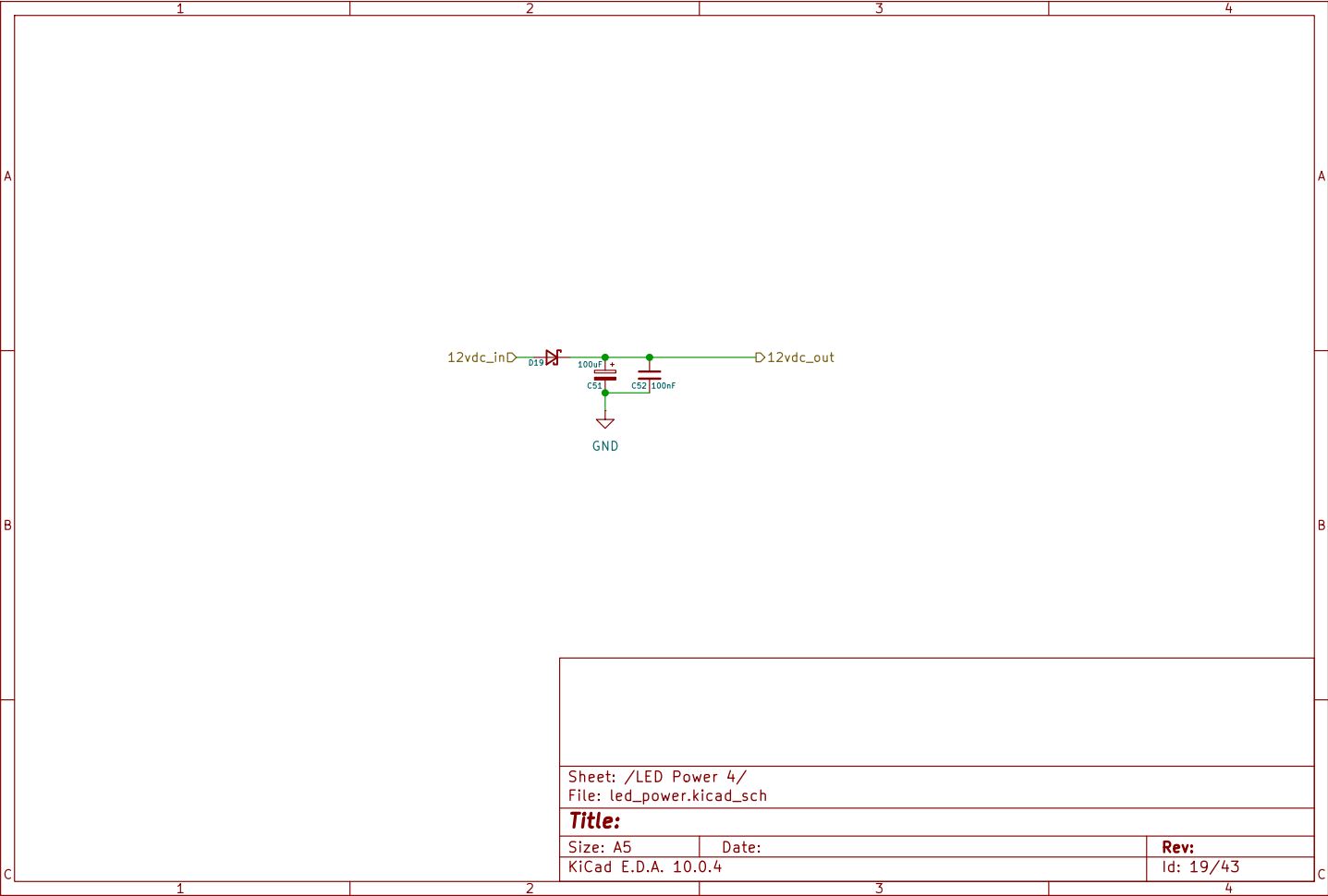
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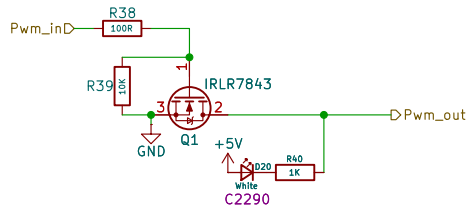
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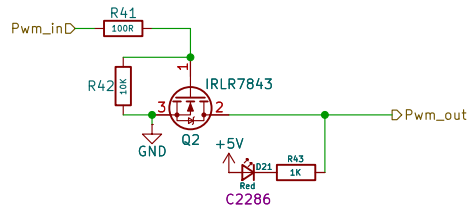
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Size: A5
KiCad E.D.A. 10.0.4

Date:

Rev:
Id: 20/43



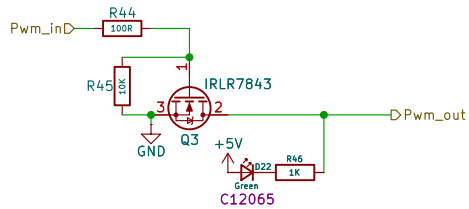
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KiCad E.D.A. 10.0.4

Date:

Rev:
Id: 21/43



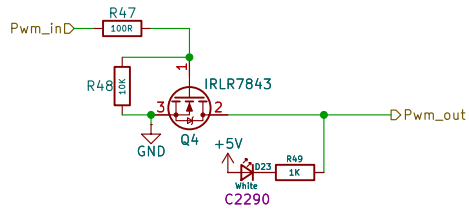
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Date:

Rev:
Id: 23/43



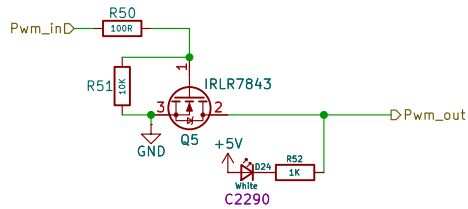
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Date:

Rev:
Id: 24/43



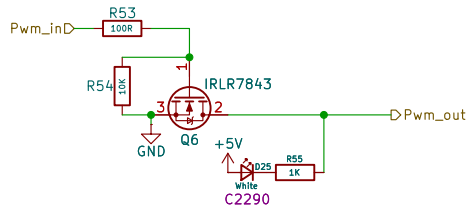
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Date:

Rev:
Id: 25/43



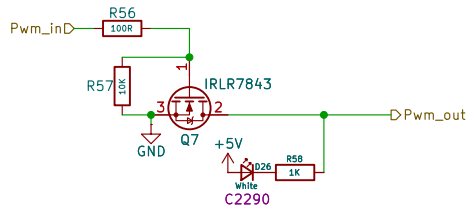
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Date:

Rev:
Id: 26/43



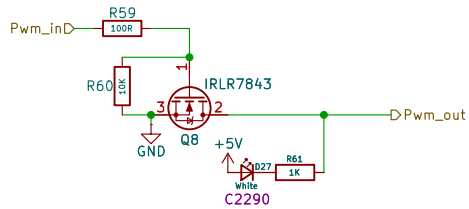
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Date:

Rev:
Id: 27/43



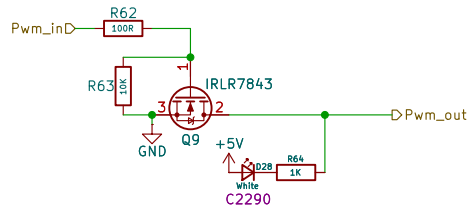
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Date:

Rev:
Id: 28/43



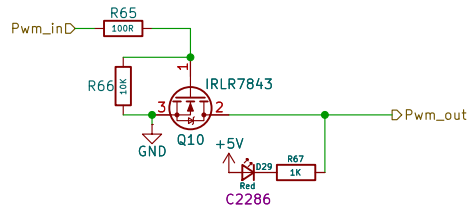
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Date:

Rev:
Id: 29/43



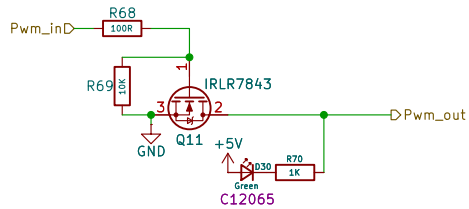
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Title:

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KiCad E.D.A. 10.0.4

Date:

Rev:
Id: 30/43



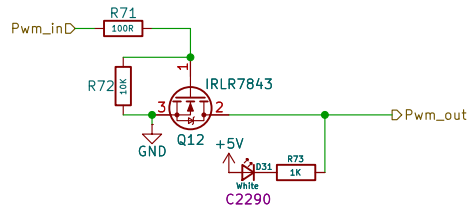
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Date:

Rev:
Id: 31/43



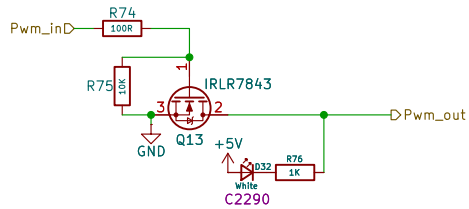
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Title:

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Date:

Rev:
Id: 32/43



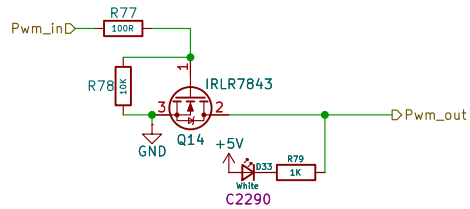
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Title:

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KiCad E.D.A. 10.0.4

Date:

Rev:
Id: 33/43



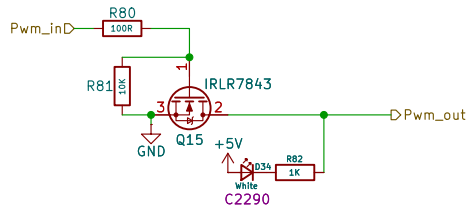
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Title:

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Date:

Rev:
Id: 34/43



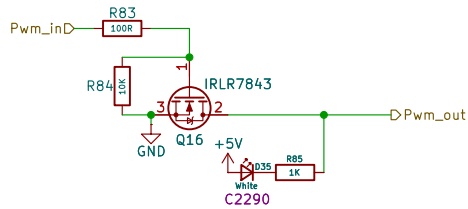
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Title:

Size: A5
KiCad E.D.A. 10.0.4

Date:

Rev:
Id: 35/43



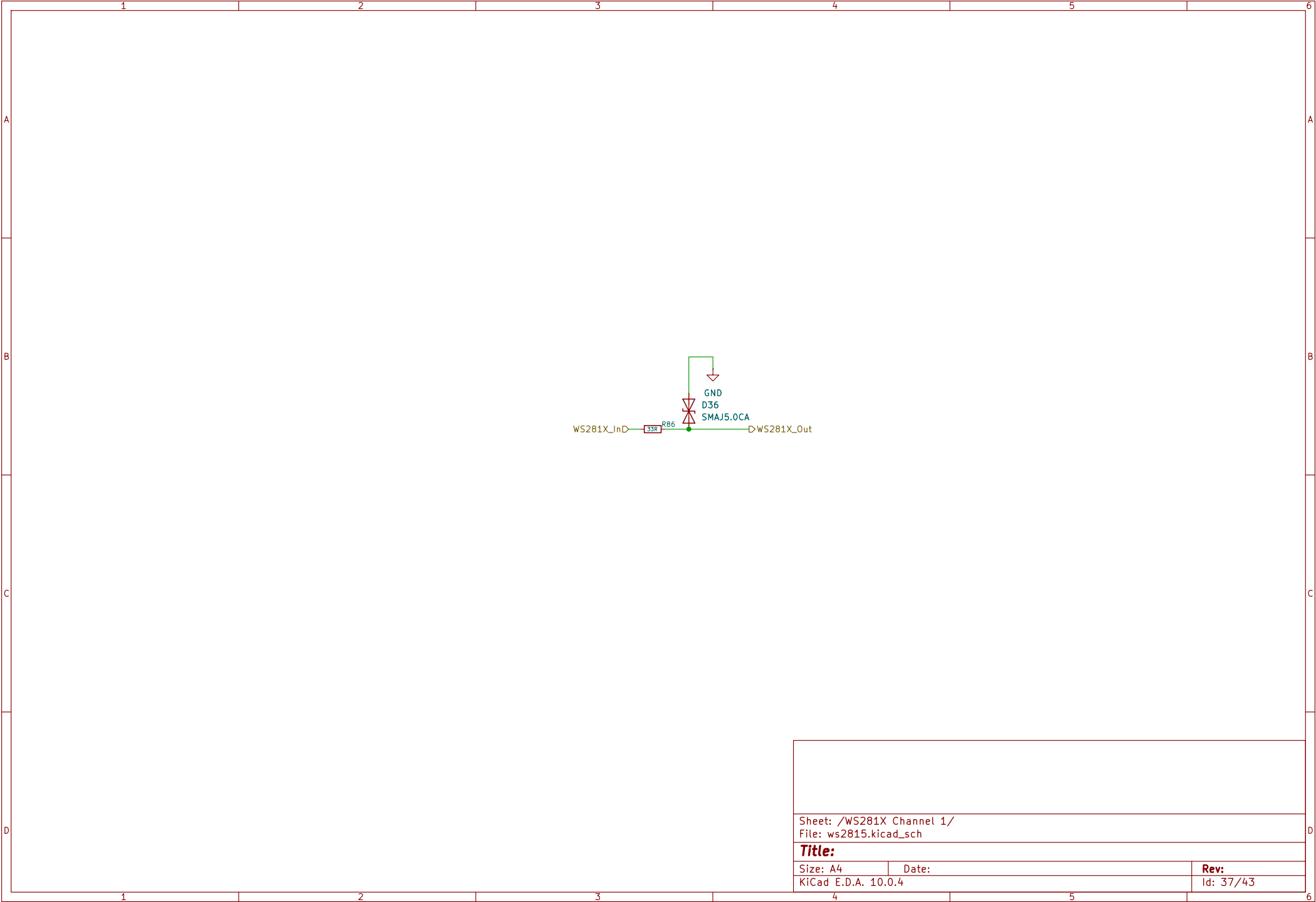
Sheet: /LED Channel 16/
File: led_channel.kicad_sch

Title:

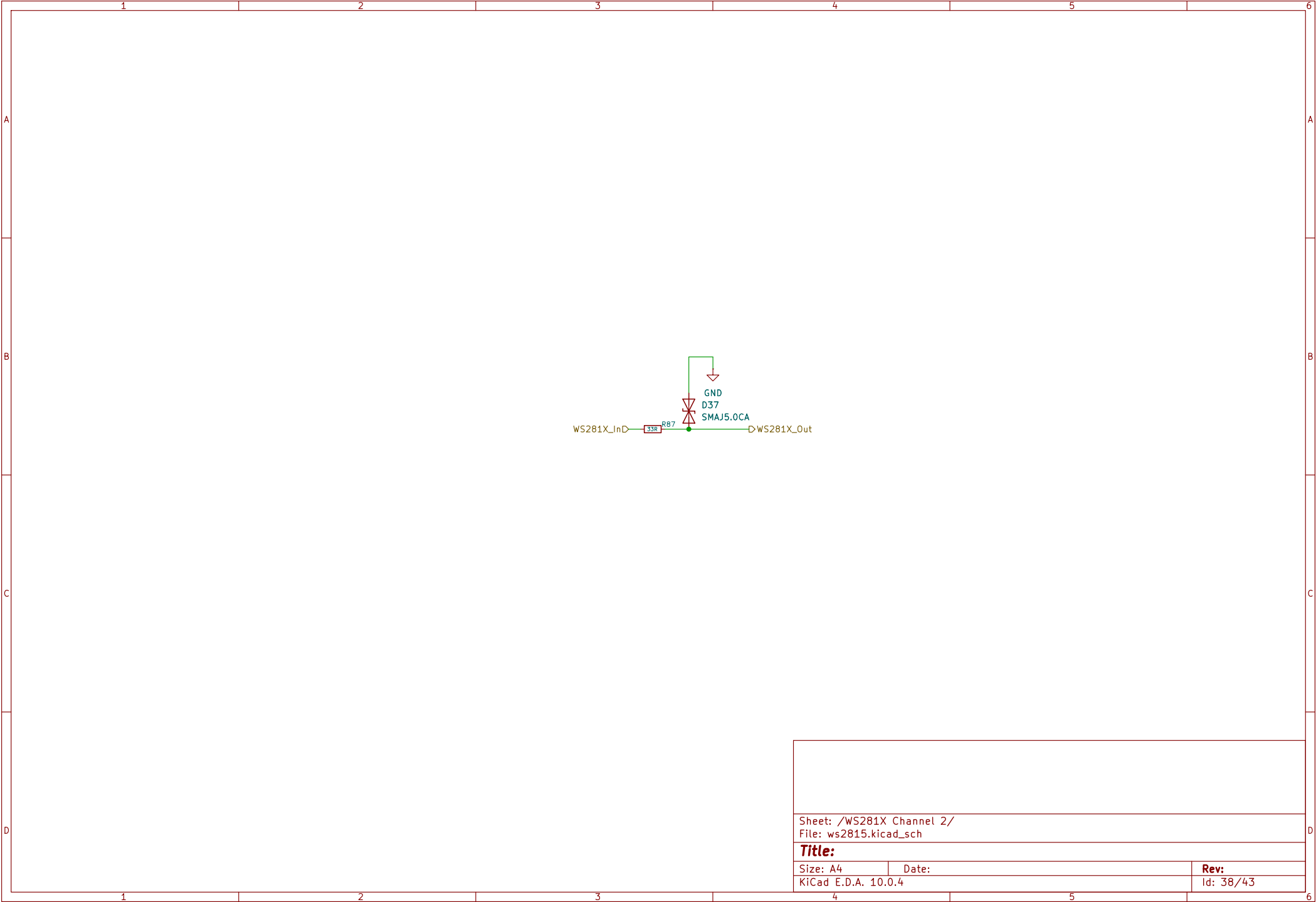
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Date:

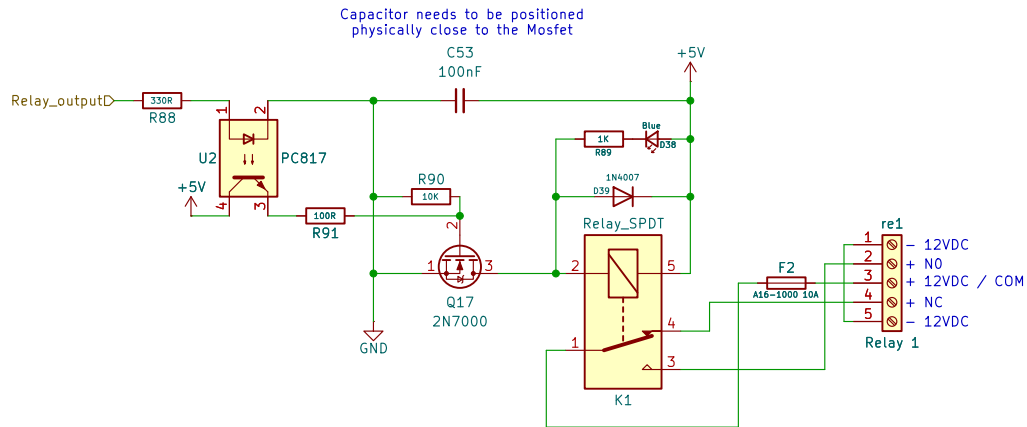
Rev:
Id: 36/43



Sheet: /WS281X Channel 1/ File: ws2815.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 37/43



Sheet: /WS281X Channel 2/ File: ws2815.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.4	Id: 38/43	



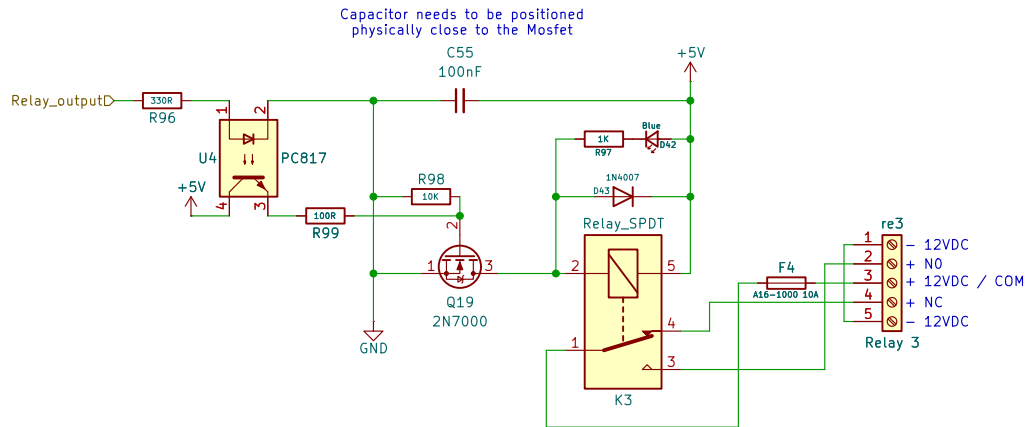
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File: relay_channel.kicad_sch

Title:

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KiCad E.D.A. 10.0.4

Date:

Rev:
Id: 39/43



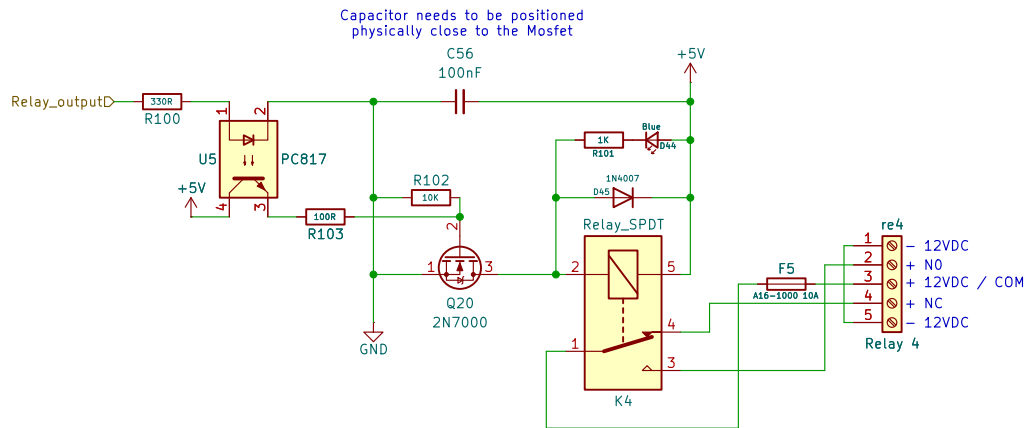
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Date:

Rev:
Id: 41/43



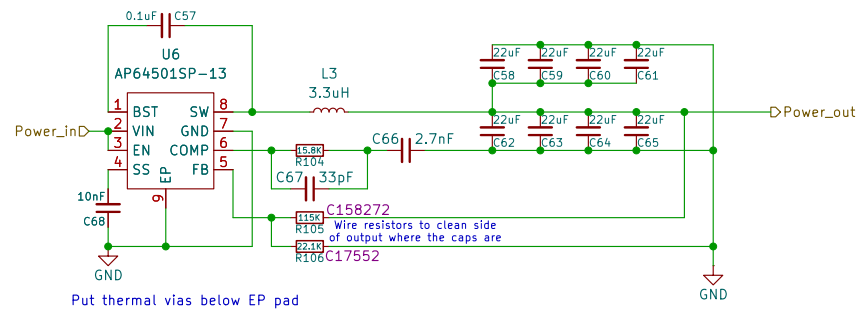
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Title:

Size: A5
KiCad E.D.A. 10.0.4

Date:

Rev:
Id: 42/43



Sheet: /Power - 5V/
File: power.kicad_sch

Title:

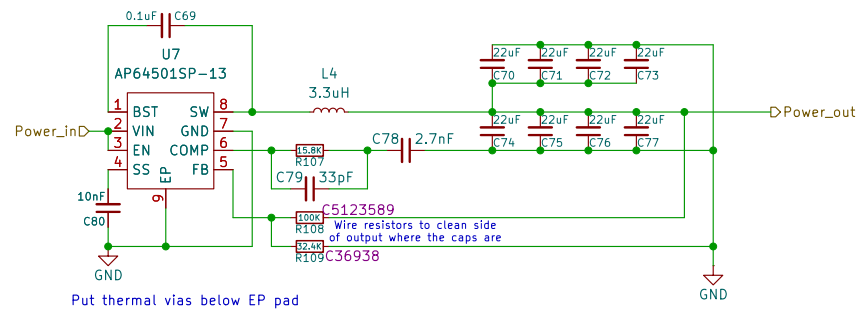
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Date:

KiCad E.D.A. 10.0.4

Rev:

Id: 43/43



Sheet: /Power - 3V3/
File: power.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 10.0.4

Rev:

Id: 44/43